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Prof. Jonas Ries
Max Perutz Labs
University of Vienna / Medical University of Vienna
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Re: PhD Position in Super-Resolution Microscopy (Ref. 5654) — late application

Dear Prof. Ries,

I am writing to express interest in the predoctoral position (Ref. 5654) in the Ries lab. I am aware the advertised deadline has passed and that the position formally requires passage through the Vienna BioCenter International PhD programme recruitment — I am reaching out directly in case the position remains open or in case you have a related opening. If neither applies now, I would welcome the opportunity to discuss the VBC selection process or future openings.

The reason I am writing despite the timing is that the lab's research programme — developing computational methods for single-molecule localization microscopy and fitting super-resolution data to geometric structural models — is precisely where my own independent research has been converging, from a different starting point.

Geometric model fitting from sparse localization data. The central computational problem in SMLM is reconstructing the geometry of a protein complex from a sparse, noisy cloud of single-molecule positions. Verum addresses the analogous problem in a different domain: a vehicle as a 20-node oscillatory circuit graph, where the complete internal state must be reconstructed from observations at 33% of nodes (5 out of 15 surface-accessible subsystems). The trajectory completion operator $T = P \circ B \circ K$ is a Banach contraction with Lipschitz constant $\lambda \in [0.3, 0.7]$, guaranteeing unique fixed-point convergence to the full system state; validated on real Formula One telemetry data (convergence residual 7.2×10^{-9}). This maps directly onto the LocMoFit problem: given a localization cloud, recover the underlying geometric model parameters. The circuit graph topology provides the structural prior; the contraction mapping provides the convergence guarantee. The same mathematical architecture — Kirchhoff propagation from boundary conditions, backward Viterbi inference, thermodynamic feasibility projection — applies when the “nodes” are protein subunits and the “observations” are fluorophore positions in a MINFLUX or STORM dataset.

Simulator-based inference and surrogate models. The lab uses simulator-based (likelihood-free) inference for fluorophore localisation in 3D. This is exactly the application class that my Purpose framework and homo-veloce surrogate model system address. The

central result — the **Backward Training Principle** — establishes that a surrogate trained at the hardest single-objective instance (the *pure-intent regime*) is competent across every less-constrained sub-regime by feasibility inclusion, not generalisation. For SMLM simulator-based inference, the pure-intent regime is the highest-SNR, lowest-density acquisition condition (one fluorophore per diffraction volume, maximum photon budget) — the regime from which all realistic acquisition conditions are feasibility restrictions. Training the surrogate there eliminates the need to sample the full acquisition parameter space. A Forward Training Collapse Theorem establishes a provably positive VC failure-gap for the conventional easy-to-hard approach; a Sample Complexity Theorem gives $(N + 1) \times$ savings for N -layer constraint cascades. The homo-veloce system demonstrates the pattern in practice: Gaussian process and graph neural network physics solvers distilled into lightweight neural networks (1D CNNs, LSTMs) for real-time inference — the same pattern the lab’s deep learning localisation pipeline implements.

Multi-modal microscopy data integration. Helicopter is a multi-modal microscopy analysis framework that combines fluorescence, brightfield, phase-contrast, and spectral imaging modalities through sequential constraint satisfaction. Each physical measurement is treated as an independent exclusion operator rather than as a signal to be averaged: the framework reduces structural ambiguity from $\sim 10^{60}$ possible molecular configurations to unique solutions without depending on the resolution limit of any single modality. For structural cell biology applications combining SMLM with cryo-EM or confocal reference channels — the kind of multi-scale structural integration the lab’s work on clathrin-mediated endocytosis and motor protein stepping requires — this is the relevant architecture: each modality contributes independent geometrical constraints, and the intersection of those constraints determines the structure.

First-principles fluorophore spectroscopy. The Borgia framework derives spectroscopic properties of molecules — electronic transitions, absorption, emission, and vibrational modes — from bounded phase space geometry alone, with zero empirical parameters. The Triple Equivalence Theorem equates oscillatory dynamics, categorical state enumeration, and partition operations; a GPU fragment shader running the partition calculation is formally equivalent to a physical spectrometer. Validated against NIST for nine elements and six molecules with self-consistency errors below 1.63%. For rational fluorophore selection in two-colour or three-colour SMLM, the ability to compute excitation and emission spectra, extinction coefficients, and Stokes shifts from molecular structure rather than from empirical measurements enables screening of candidate labels computationally before synthesis or purchase. The same framework — Beer-Lambert absorption derived from Kirchhoff circuit laws (proved in the Lagrangian platform, Resonant Stack theorem, $T = 1/e = 0.3679$ for both) — provides a first-principles model of light propagation through biological tissue for thick-sample PSF characterisation and aberration correction in 3D SMLM.

Scientific data querying and localization database search. A complementary framework — the PDSVM (Purpose-Driven Shader Virtual Machine) — addresses large-scale scientific database search via bounded oscillatory systems. Every stable physical object is described by three S-entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$ (knowledge, temporal, and evolution entropy); every query is classified into one of three GPU shader primitive classes (proved exhaustive by the Completeness Theorem); and retrieval via a ternary trie over Morton-ordered S-entropy coordinates runs in $O(k)$ time independent of database size N ($373\times$ speedup at $N = 10^5$, < 0.01 ms per iteration on a consumer GPU, WebGPU

browser-deployable). The stable slice T^* is the fixed point of the probe operator $\Pi_P(T)$ — the set of data states whose S-entropy coordinates are closest to a given purpose point P on the Fisher-metric manifold. This maps exactly onto the LocMoFit problem structure: the localization cloud is the data set; the target geometric model is the purpose point P ; and the stable slice is the protein complex geometry satisfying the forcing conditions (structural constraints derivable without statistical averaging). For a lab that builds and extends LocMoFit, the trie gives $O(k)$ localization database search independent of cloud size, and the stable slice formalism provides a first-principles account of which geometric solutions are structurally necessary rather than merely statistically likely. 10/10 validation experiments all pass (including generative capability: 1000/1000 instances satisfy forcing conditions within $\varepsilon = 0.05$ of the purpose point).

Background. My MSc in Computational Biology (Universität Tübingen) and doctoral research in Computational Lipidomics at TUM give me the biological and computational science foundation. My primary development environment is Python (PyTorch, scikit-learn, Ray) and Rust for performance-critical components. I have not operated STORM or MINFLUX hardware — I am aware that the PhD programme provides that training — but the computational methods I have developed are directly applicable to the data analysis and method development problems the lab addresses. I am legally resident in Germany with full work authorisation and can commute to Vienna or relocate if needed. English is native; German is conversational.

I recognise this is a late contact and appreciate your time in considering it.

Yours sincerely,

Kundai Farai Sachikonye